

ECE 473 – Introduction to Cloud Computing and Cloud Native Systems
ECE 573 – Cloud Computing and Cloud Native Systems
Fall 2026

Instructor: Professor Jia Wang

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Prerequisites: ECE 242 Digital Computers and Computing

Class Time and Location: Mon./Wed. 1:50 PM - 3:05 PM,
Mon. Hermann Hall 005, Wed. IIT Tower 6C7-1

Class Home Page: <https://wngjia.github.io/ece573-web/>

Required Textbook:

- “Cloud Native Go: Building Reliable Services in Unreliable Environments”
Matthew Titmus, O’Reilly Media, 2021, ISBN-13 978-1492076339
- Plus additional research papers

Computer and Network Requirements:

- A recent Windows computer with 4 CPU cores, 16GB memory, and 512GB SSD; or access to a x64 Ubuntu server with 4 CPU cores, 16GB memory, and 100GB storage.
 - We are not able to support ARM-based computers like Apple MacBooks and Raspberry Pi’s, as well as computers that are more than 5 years old.
- Internet access to common code and package repositories like GitHub and Docker Hub.
- (Recommended) Subscription to coding agent plans like Claude Code or Codex.

Course Descriptions:

ECE 473: This course introduces students to cloud native systems that build on top of the cloud computing architecture to provide scalable services in dynamic environments. Key topics covered include virtualization and containerization, distributed database systems, communication mechanisms, batch and stream processing, resource management, consensus, security, and system design techniques for scalability, resilience, manageability, and observability. Course projects will provide hands-on experiences with state-of-the-art languages, libraries, and tools.

ECE 573: This course builds on foundational cloud computing and cloud native concepts through a deeper treatment of distributed systems. Key topics covered include distributed database systems, consensus, resource management, communication mechanisms, batch and stream processing, and cloud native system design. Course projects will provide hands-on experiences with state-of-the-art languages, libraries, and tools, and include advanced work in web security and chaos engineering.

Grading:

- 5 Homeworks: 10 points total.
- 7 Projects: 120 points total.
- ECE 473: $A \geq 90$ / $B \geq 80$ / $C \geq 60$ / D (undergraduate only) ≥ 55 .
- ECE 573: $A \geq 115$ / $B \geq 100$ / $C \geq 80$.

Homework and Project Policy: Late homeworks and project reports will not be graded. Discussions on homeworks and projects are encouraged, but copying will call for disciplinary action.

Lecture Schedule (tentative):

No.	Date	Topic	Chapters	HW Out	Project Due
1, 2	8/17, 8/19	Introduction	1,6	HW #1	
3, 4	8/24, 8/26	Go Introduction	2,3		
5, 6	8/31, 9/2	Virtualization, Containerization		HW #2	
7	9/7, 9/9	RESTful Services	5		
8, 9	9/14, 9/16	Cloud Native Patterns	4	HW #3	PRJ #1
10,11	9/21, 9/23	Web Security			
12,13	9/28, 9/30	Cloud Storage			PRJ #2
14,15	10/5, 10/7	Distributed Database Systems			
16	10/12, 10/14	Scalability	7		PRJ #3
17,18	10/19, 10/21	Resource Management I		HW #4	
19,20	10/26, 10/28	Resource Management II			PRJ #4
21,22	11/2, 11/4	Communication Mechanisms	8	HW #5	
23,24	11/9, 11/11	Resilience and Chaos Engineering	9		PRJ #5
25,26	11/16, 11/18	Manageability and Observability	10,11		
27	11/23, 11/25	Batch and Stream Processing			
28,29	11/30,12/2	Consensus			PRJ #6
	12/7–12/11	No Final Exam			PRJ #7

ECE 473 Course Objectives (ABET)

After completing this course, you should be able to:

1. Identify scenarios where cloud computing applies.
2. Understand virtualization and containerization techniques.
3. Explain common storage, communication, and computing patterns in cloud.
4. Describe cloud native system design techniques for security, scalability, resilience, manageability, and observability.
5. Utilize open-source software to build cloud native systems.

AI Usage Policy: The instructor may use AI tools to develop course content and please be assured that the instructor maintains final oversight, review, and ultimate responsibility for all course materials.

You are encouraged to use AI tools as part of learning and project development in this course with proper acknowledgment. Nevertheless, you are ultimately responsible for the content you submit, regardless of its source.

ADA Statement: Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources and make an appointment to speak with me as soon as possible. The Center for Disability Resources is located in the Life Sciences Building, room 218, 312-567-5744 or disabilities@iit.edu.

Sexual Harassment and Discrimination Information: Illinois Tech prohibits sexual harassment by any member of the community. This includes harassment among students, staff, or faculty. Sexual harassment by a faculty member or teaching assistant of a student over whom they have authority can be particularly serious. Such conduct may easily create an intimidating, hostile, or offensive environment. Accordingly, Illinois Tech strives to provide for its students, faculty and staff an educational and employment environment free of sexual harassment.

Illinois Tech encourages anyone experiencing sexual harassment to speak with the Office of Title IX Compliance for information on support options and the resolution process. You can report sexual harassment electronically at iit.edu/incidentreport or contact either the Title IX Coordinator, Virginia Foster at foster@illinoistech.edu, the Title IX Specialist, Ryley Andrews at randrews2@illinoistech.edu

To speak with someone confidentially, contact Illinois Tech's Confidential Advisor at (773) 275 8340 or a licensed practitioner in Illinois Tech's Student Health and Wellness Center by emailing student.health@illinoistech.edu or calling (312)567-7550. For a comprehensive list of resources, visit the Office of Title IX Compliance website at <https://www.iit.edu/title-ix/resources>.

University Resources for Students' Rights, Responsibilities, and Related Policies:

- The Center for Disability Resources (CDR) Rights and Responsibilities
<https://www.iit.edu/cdr/faqs/rights-and-responsibilities>
- The Family Educational Rights and Privacy Act (FERPA)
<https://www.iit.edu/registrar/students-and-alumni/ferpa>
- Code of Conduct (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/code-conduct>
- Policies, Procedures, and Regulations (from Student Handbook)
<https://www.iit.edu/student-affairs/student-handbook/fine-print/policies-regulations-and-procedures>